

1 Introduction

Our notation:

- Random variable X_i with value x_i and possible values from $\mathcal{X}$,
- Random vector $\mathbf{X}$ with range $\mathcal{X}$ and some assignment $\mathbf{x}$,
- Set $\mathcal{A}$, and a measure $\mathbb{M}$.

2 Making Decisions under Uncertainty

3 Convergent Prior

Definition 3.1 (Probabilistic Model). We define a *probabilistic model* $\mathcal{M}$ as a pair $(\mathcal{S}, \mathcal{P})$, where $\mathcal{S}$ is a sample space, and $\mathcal{P}$ is a set of probability measures on $\mathcal{S}$.

Definition 3.2 (Divergence Measure). Let $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \dots\}$ be a set of pairs of discrete random variables in a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$. We define the *divergence measure of $P \in \mathcal{P}$ for $\mathcal{X}$* as follows:

$$\mathbb{D}_{\mathcal{X}} [P] = \max_{Q \in \mathcal{P}} \mathbb{E}_Q \left[\sum_i \mathbb{D} (Q_{X_i|e_i} \parallel P_{X_i|e_i}) \right], \quad (1)$$

where $\mathbb{E}_Q [\cdot]$ denotes expectation relative to Q and each $\mathbb{D} (Q_{X_i|e_i} \parallel P_{X_i|e_i})$ is the Kullback-Leibler divergence between $P(X_i|e_i)$ and $Q(X_i|e_i)$:

$$\mathbb{D} (Q_{X_i|e_i} \parallel P_{X_i|e_i}) = \sum_{x_i} Q(x_i | e_i) \ln \frac{Q(x_i | e_i)}{P(x_i | e_i)}.$$

Moreover, we define *the central distribution for $\mathcal{X}$* to be a distribution $C_{\mathcal{X}} \in \mathcal{P}$ which minimizes the divergence measure for $\mathcal{X}$:

$$C_{\mathcal{X}} = \arg \min_{P \in \mathcal{P}} \mathbb{D}_{\mathcal{X}} [P]. \quad (2)$$

Proposition. Using the linearity of expectation, we have:

$$\mathbb{D}_{\mathcal{X}} [P] = \max_{Q \in \mathcal{P}} \sum_i \sum_{x_i, e_i} Q(x_i, e_i) \ln \frac{Q(x_i | e_i)}{P(x_i | e_i)}.$$

Definition 3.3 (Parametric Model). Let Θ be a random vector. A parametric model is a probabilistic model $\mathcal{M}_\theta \equiv (\mathcal{S}, \mathcal{P})$, where for every $P \in \mathcal{P}$ there is a prior probability p , such that for every $s \in \mathcal{S}$ we have:

$$P(s) = \int \pi(s \mid \theta) p(\theta) d\theta, \quad (3)$$

while π is some known probability distribution.

Lemma 1. Let $\mathcal{M}_\theta \equiv (\mathcal{S}, \mathcal{P})$ be a parametric model, and $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \dots\}$ be a set of discrete random variable pairs in $\mathcal{M}_\theta$. For every $P \in \mathcal{P}$ we have:

$$\mathbb{D}_{\mathcal{X}}[P] \leq \max_{\theta \in \Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel P_{X_i|e_i}) \mid \theta \right] \quad (4)$$

$$= \max_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)}. \quad (5)$$

Proof. Let R be an arbitrary distribution in $\mathcal{M}_\theta$, since the Kullback-Leibler divergence is always positive, its expected value is also positive, and for every parameter value θ we have:

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{R(x_i \mid e_i)} \geq 0.$$

Using this, we can derive the following inequality for every distribution P and every parameter θ :

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)}.$$

Since this inequality holds for any θ , the maximum of the right-hand side is also greater than or equal to the maximum of the left-hand side w.r.t θ :

$$\max_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \max_{\theta \in \Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel P_{X_i|e_i}) \mid \theta \right]. \quad (6)$$

$\mathcal{M}_\theta$ is a parametric model and $R(x_i, e_i) = \int \pi(x_i, e_i \mid \theta) r(\theta) d\theta$, while $r(\theta)$ is a probability distribution. Obviously for every θ we have:

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \max_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)}.$$

Since $r(\theta)$ is a distribution and is non-negative, we can multiply this inequality by $r(\theta)$ and integrate. Notice that the right hand side is not a function of θ and it will remain unchanged because $\int r(\theta)d\theta = 1$:

$$\mathbb{E}_R \left[\sum_i \mathbb{D}(R_{X_i|e_i} \parallel P_{X_i|e_i}) \right] \leq \max_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)}.$$

Now by Inequality 6, for *every* distribution P and R we conclude:

$$\mathbb{E}_R \left[\sum_i \mathbb{D}(R_{X_i|e_i} \parallel P_{X_i|e_i}) \right] \leq \max_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)},$$

which proves the theorem. $\square$

Theorem 2. Let $\mathcal{M}_\theta \equiv (\mathcal{S}, \mathcal{P})$ be a parametric model and $\mathcal{X}$ be a set of pairs of discrete random variables in $\mathcal{M}_\theta$. If for every value of θ there is a distribution $R \in \mathcal{P}$ such that $R(s) = \pi(s \mid \theta)$ for every $s \in \mathcal{S}$, then for any $P \in \mathcal{P}$ we have:

$$\mathbb{D}_{\mathcal{X}}[P] = \max_{\theta \in \Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel P_{X_i|e_i}) \mid \theta \right]. \quad (7)$$

Proof. For an arbitrary distribution $P \in \mathcal{P}$, we let:

$$\theta^* = \arg \max_{\theta \in \Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel P_{X_i|e_i}) \mid \theta \right].$$

In $\mathcal{M}_\theta$ there should be a distribution $R \in \mathcal{P}$ that equals $\pi(\cdot \mid \theta^*)$, and we have:

$$\mathbb{E}_R \left[\sum_i \mathbb{D}(R_{X_i|e_i} \parallel P_{X_i|e_i}) \right] = \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta^*} \parallel P_{X_i|e_i}) \mid \theta^* \right].$$

On the other hand, we see:

$$\max_{Q \in \mathcal{P}} \mathbb{E}_Q \left[\sum_i \mathbb{D}(Q_{X_i|e_i} \parallel P_{X_i|e_i}) \right] \geq \mathbb{E}_R \left[\sum_i \mathbb{D}(R_{X_i|e_i} \parallel P_{X_i|e_i}) \right],$$

Thus:

$$\mathbb{D}_{\mathcal{X}}[P] \geq \max_{\theta \in \Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel P_{X_i|e_i}) \mid \theta \right].$$

This and Inequality 5 proves the theorem. $\square$

Theorem 3 (Minimax theorem [1]). *Let $\mathcal{X} \subset \mathbb{R}^n$ and $\mathcal{Y} \subset \mathbb{R}^m$ be compact convex sets. If $f : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ is a continuous function that is concave-convex, i.e.*

- $f(\cdot, y) : \mathcal{X} \rightarrow \mathbb{R}$ is concave for fixed y , and
- $f(x, \cdot) : \mathcal{Y} \rightarrow \mathbb{R}$ is convex for fixed x .

Then we have that:

$$\max_{x \in \mathcal{X}} \min_{y \in \mathcal{Y}} f(x, y) = \min_{y \in \mathcal{Y}} \max_{x \in \mathcal{X}} f(x, y). \quad (8)$$

Theorem 4. *Consider a generic parametric model $\mathcal{M}_\theta \equiv (\mathcal{S}, \mathcal{P})$, and let $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \dots\}$ be a set of pairs of discrete random variables in $\mathcal{M}_\theta$. Assume that for every $Q_1, Q_2 \in \mathcal{P}$ and any real number $0 \leq \alpha \leq 1$, there is a distribution $Q^* \in \mathcal{P}$, such that for every x_i, e_i, i we have:*

$$Q^*(x_i | e_i) = \alpha Q_1(x_i | e_i) + (1 - \alpha) Q_2(x_i | e_i). \quad (9)$$

In this model, the central distribution for $\mathcal{X}$ can be obtained by:

$$C_{\mathcal{X}} = \arg \max_{P \in \mathcal{P}} \int \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel P_{X_i|e_i}) \middle| \theta \right] p(\theta) d\theta \quad (10)$$

$$= \arg \max_{P \in \mathcal{P}} \int \sum_i \sum_{x_i, e_i} P(x_i, e_i, \theta) \ln \frac{\pi(x_i | e_i, \theta)}{P(x_i | e_i)} d\theta, \quad (11)$$

where $P(x_i, e_i, \theta) = \pi(x_i, e_i | \theta) p(\theta)$.

Proof. We prove this theorem using the Minimax theorem. For that, we define a functional as follows:

$$f(r, P_{X_i|e_i}) = \int \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel P_{X_i|e_i}) \middle| \theta \right] r(\theta) d\theta,$$

where $r(\theta)$ is an arbitrary probability distribution, and $P \in \mathcal{P}$. Since $r(\theta)$ can be any distribution, maximizing with respect to r is like maximizing with respect to θ . Thus we have:

$$C_{\mathcal{X}} = \arg \min_{P \in \mathcal{P}} \max_r f(r, P_{X_i|e_i}).$$

The domain of r is the set of all probability distributions which is a compact convex set. Moreover, by Equation 9 the set of probability distributions for $X_i|E_i$ in $\mathcal{M}_\theta$ is also a compact convex set. Therefore, the domain of $f(r, P_{X_i|e_i})$ is a compact convex set.

Obviously, $f(r, \cdot)$ is a linear function, which is both concave and convex. On the other hand, if we let $R(x_i, e_i) = \int \pi(x_i, e_i | \theta) r(\theta) d\theta$ for $f(\cdot, P_{X_i|e_i})$ we have:

$$f(\cdot, P_{X_i|e_i}) = c - \sum_i \sum_{x_i, e_i} R(x_i, e_i) \ln P(x_i | e_i). \quad (12)$$

If we calculate the Hessian matrix of $f(\cdot, P_{X_i|e_i})$, we observe that the matrix is diagonal, and each element of the main diagonal is $-\frac{R(x_i|e_i)}{P(x_i|e_i)^2}$. Thus, the hessian matrix of $f(\cdot, P_{X_i|e_i})$ is negative semi-definite and $f(\cdot, P_{X_i|e_i})$ is a concave function. Now, by Theorem 3 we have:

$$\min_{P \in \mathcal{P}} \max_r f(r, P_{X_i|e_i}) = \max_r \min_{P \in \mathcal{P}} f(r, P_{X_i|e_i}). \quad (13)$$

By Equation 12 we can see that for an arbitrary fixed r we have:

$$\min_{P \in \mathcal{P}} f(r, P_{X_i|e_i}) = \max_{P \in \mathcal{P}} \sum_i \sum_{x_i, e_i} R(x_i, e_i) \ln P(x_i | e_i).$$

We know that the Kullback–Leibler divergence, and therefore its expected value, is non-negative:

$$\mathbb{E}_R \left[\sum_i \mathbb{D}(R_{X_i|e_i} \parallel P_{X_i|e_i}) \right] \geq 0.$$

By simple algebra we get:

$$\sum_i \sum_{x_i, e_i} R(x_i, e_i) \ln P(x_i | e_i) \leq \sum_i \sum_{x_i, e_i} R(x_i, e_i) \ln R(x_i | e_i).$$

Remember that $\mathcal{M}_\theta$ is a generic parametric model, so $R \in \mathcal{P}$ and we can reach this upper bound by letting $P = R$.

$$\min_{P \in \mathcal{P}} f(r, P_{X_i|e_i}) = \int \sum_i \sum_{x_i, e_i} R(x_i, e_i, \theta) \ln \frac{\pi(x_i | e_i, \theta)}{R(x_i | e_i)} d\theta.$$

By Equation 13 we conclude:

$$C_{\mathcal{X}} = \arg \max_{R \in \mathcal{R}} \int \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel R_{X_i|e_i}) \mid \theta \right] r(\theta) d\theta.$$

□

References

- [1] Ding-Zhu Du. *Minimax and Its Applications*, pages 339–367. Springer US, Boston, MA, 1995.